

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

Artificial Intelligence & Machine Learning – Task 1

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Dataset: California Housing Dataset

Algorithm: Linear Regression

Objective

Develop and evaluate a Linear Regression model to predict house prices using the California Housing dataset while demonstrating the complete machine learning workflow.

Introduction

Linear Regression is a supervised machine learning algorithm used to predict continuous values. In this project it is applied to estimate house prices from housing-related features.

Dataset Description

The California Housing dataset contains numerical attributes including Median Income, House Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude and Longitude. The target variable is House Price.

```
from sklearn.datasets import fetch_california_housing
housing = fetch_california_housing()

df = pd.DataFrame(housing.data, columns=housing.feature_names)
df["HousePrice"] = housing.target

print("Dataset loaded successfully!")
df.head()
```

Dataset loaded successfully!

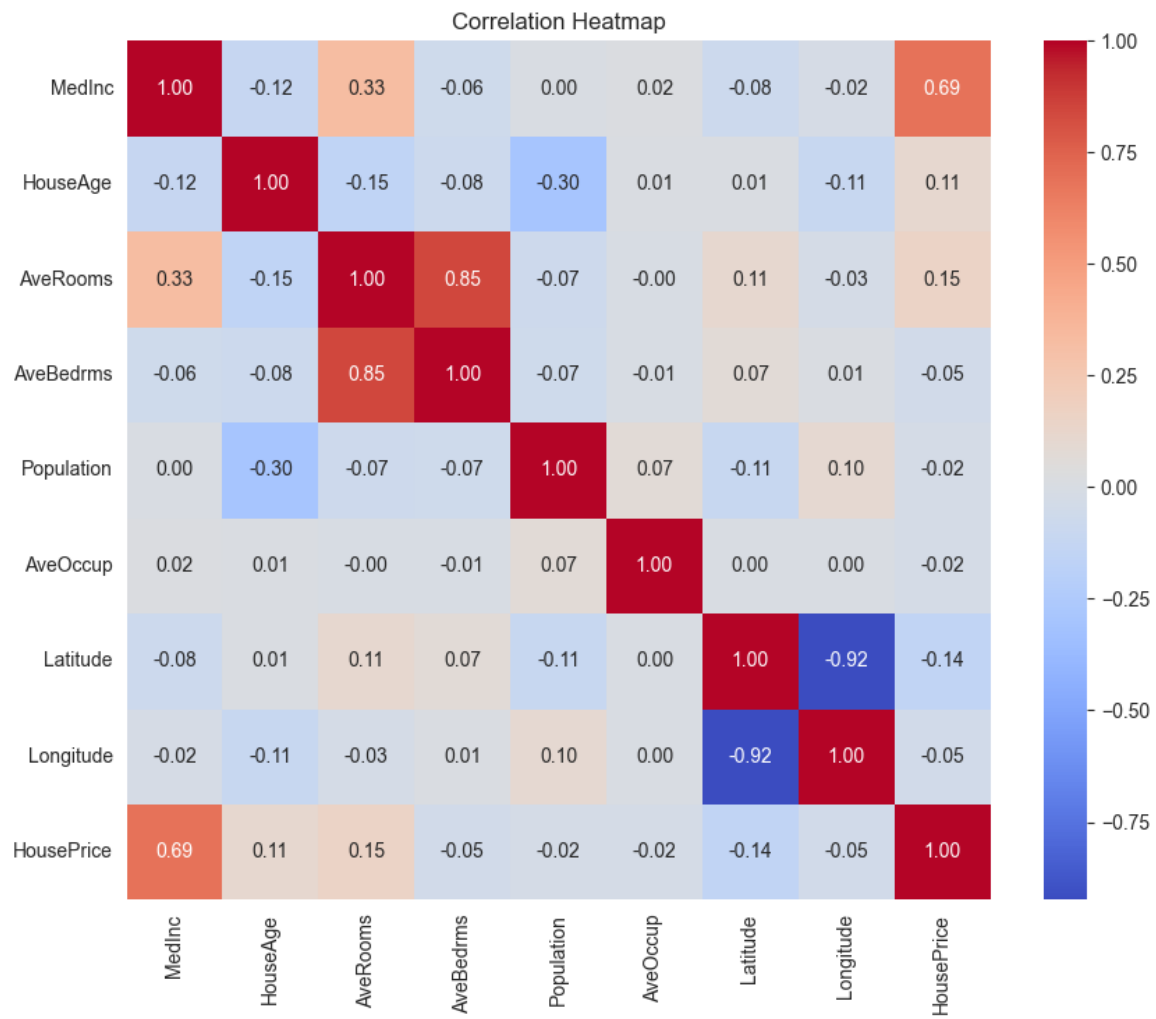
	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	HousePrice
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Methodology

- Import libraries
- Load dataset
- Perform Exploratory Data Analysis (EDA)
- Check missing values and statistics
- Create visualizations
- Split data into training (80%) and testing (20%)
- Train Linear Regression model
- Predict house prices
- Evaluate using MAE, MSE, RMSE and R² Score

Exploratory Data Analysis

EDA included dataset shape, column names, data types, missing values, descriptive statistics, histograms, correlation heatmap, and scatter plots.



Model Evaluation

Insert the metric values from your notebook:

```
mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)

# Print the results
print("Model Evaluation Results")
print("-" * 35)

print(f"Mean Absolute Error (MAE) : {mae:.4f}")
print(f"Mean Squared Error (MSE) : {mse:.4f}")
print(f"Root Mean Squared Error (RMSE) : {rmse:.4f}")
print(f"R2 Score : {r2:.4f}")
```

```
Model Evaluation Results
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Mean Absolute Error (MAE) : 0.5332
Mean Squared Error (MSE) : 0.5559
Root Mean Squared Error (RMSE) : 0.7456
R2 Score : 0.5758
```

Results

The model successfully learned relationships between housing features and prices. Predicted values were reasonably close to actual values, demonstrating the usefulness of Linear Regression.

Future Improvements

Use feature engineering, feature scaling, cross-validation, hyperparameter tuning, and advanced regression algorithms such as Random Forest or XGBoost.

Conclusion

The project successfully implemented a Linear Regression model using the California Housing dataset. The workflow included data loading, EDA, visualization, model training, prediction, and evaluation. This project provided practical experience with supervised machine learning.